Statistical Structured Prediction

3. Making Prediction: Decoding and Inference

3.1. Linear models:

3.1.1. Discriminative models: $\Rightarrow$ CRFs

- ➢ Decoding: **Viterbi,**
- ➢ Inference: **Fordward**

3.1.2. Generative models: $\Rightarrow$ HMMs

- ➢ Decoding: **Viterbi,**
- ➢ Inference: **Fordward**

3.2. Non-linear (generative) models: $\Rightarrow$ PCFGs

- ➢ Decoding: probability of most likely parse tree: **Viterbi**
- ➢ Inference: probability of a string: **Inside**
- ➢ Weaknesses of PCFGs as parsing models

Conditional Random Fields

Given $x = x_1, x_2, \ldots, x_T \in \mathcal{X}^*$ and $y = y_1, y_2, \ldots, y_T \in \mathcal{Y}^*$

Discriminative models: **Conditional Random Fields (CRF)**

`[Lafferty, McCallum, Pereira, 2001].`

$$p(y|x;\theta) = \frac{\exp\{\sum_{t=1}^{T}\sum_{k=1}^{K}\theta_k\, f_k(y_{t-1},\, y_t,\, x_t)\}}{Z(x;\theta)} \tag{4}$$

Where

$$Z(x;\theta) = \sum_{y'\in y^*}\exp\left\{\sum_{t=1}^{T}\sum_{k=1}^{K}\theta_k\, f_k(y'_{t-1},\, y'_t,\, x_t)\right\} \tag{5}$$

CRFs as factor graphs

We can define a transition (factor) as:

$$\Psi_t\,(y_{t-1},\, y_t,\, x_t) \stackrel{\text{def}}{=} \exp\{\sum_{k=1}^{K}\theta_k\, f_k\,(y_{t-1},\, y_t,\, x_t)\}$$

From (4) CRFs can be defined as:

$$p(y|x;\theta) = \frac{1}{Z(x;\theta)}\prod_{t=1}^{T}\Psi_t\,(y_{t-1},\, y_t,\, x_t)$$

Also from (5)

$$Z(x;\theta) = \sum_{y'\in y^*}\prod_{t=1}^{T}\Psi_t\,(y'_{t-1},\, y'_t,\, x_t)$$

CRFs as factor graphs

- ➢ Decoding: Given θ and x, predict the best output sequence for x.

$$\widehat{y} = \arg\max_y\, p(y|x;\theta) = \arg\max_y \prod_{t=1}^{T}\Psi_t\,(y_{t-1},\, y_t,\, x_t)$$

- ➢ The probability of an output sequence

$$p(y|x;\,\theta) = \frac{1}{Z(x;\,\theta)}\prod_{t=1}^{T}\Psi_t\,(y_{t-1},\, y_t,\, x_t)$$

DECODING WITH CRFS: VITERBI ALGORITHM

➢ **Definition**: Score of optimal sequence for $x_1 \dots x_t$ ending at $y_t = s \in \mathcal{Y}$

$$\gamma_t(s) \stackrel{\text{def}}{=} \max_{y_1^t;\, y_t = s} \prod_{i=1}^{t} \Psi_i\,(y_{i-1},\ y_i,\ x_i)$$

➢ **Initialization**: $\forall\, s \in \mathcal{Y}$

$$\gamma_1(s) \ = \ \Psi_1\,(y_0 = \text{null},\ y_1 = s,\ x_1)$$

➢ **Recursion**: $\forall\, t = 2 \dots T$; and $\forall\, s \in \mathcal{Y}$

$$\gamma_t(s) \ = \ \max_{s' \in \mathcal{Y}} \left\{ \ \gamma_{t-1}(s') \ \cdot \ \Psi_t\,(t_{t-1} = s',\ y_t = s,\ x_t) \ \right\}$$

➢ **Final result**: The optimal score for x is:

$$\max_s \ \gamma_T\ (s)$$

Backpointers can be used to recover the optimal sequence $\widehat{y}$

INFERENCE WITH CRFS: FORWARD ALGORITHM

➢ **Definition**: Score for $x_1 \dots x_t$ ending at $y_t = s \in \mathcal{Y}$

$$\alpha_t(s) \stackrel{\text{def}}{=} \sum_{y_1^t;\, y_t = s} \prod_{i=1}^{t} \Psi_i\,(y_{i-1},\ y_i,\ x_i)$$

➢ **Initialization**: $\forall\, s \in \mathcal{Y}$

$$\alpha_1(s) \ = \ \Psi_1\,(y_0 = \text{null},\ y_1 = s,\ x_1)$$

➢ **Recursion**: $\forall\, t = 2 \dots T$; and $\forall\, s \in \mathcal{Y}$

$$\alpha_t(s) \ = \ \sum_{s' \in \mathcal{Y}} \alpha_{t-1}(s') \ \cdot \ \Psi_t\,(y_{t-1} = s',\ y_t = s,\ x_t)$$

➢ **Final result**: The optimal score for x is:

$$\sum_s \ \alpha_T\ (s)$$

CRFS AS FACTOR GRAPHS

➢ The probability of an output sequence

$$p(y|x;\theta) \ = \ \frac{1}{Z(x;\theta)} \prod_{t=1}^{T} \Psi_t\,(y_{t-1},\ y_t,\ x_t)$$

Where we can compute $Z(x;\,\theta)$ efficiently using the forward algorithm

$$Z(x;\ \theta) \ = \ \sum_{y_1'^T} \prod_{t=1}^{T} \Psi_t(y'_{t-1},\ y'_t,\ x_t) \ = \ \sum_{s} \alpha_T(s)$$

HIDDEN MARKOV MODELS

Given $x = x_1, x_2, \dots, x_T \in \mathcal{X}^*$ and $y = y_1, y_2, \dots, y_T \in \mathcal{Y}^*$

Generative models: **Hidden Markov Models** $(\mathcal{Q}, \Sigma, \theta)$, where $\theta = (q, e)$

$$P_\theta(x_1^T, y_1^T) \ = \ \prod_{t=1}^{T} \ q(y_t|y_{t-1}) \ \cdot \ e(x_t|y_t)$$

➢ Probability of the observation

$$P_\theta(x_1^T) \ = \ \sum_{y_1^T} \ P_\theta(x_1^T, y_1^T)$$

➢ Probability of the best state sequence

$$\widehat{P}_\theta(x_1^T) \ = \ \max_{y_1^T} \ P_\theta(x_1^T, y_1^T)$$

➢ Finding the best state sequence

$$\widehat{y}_1^T \ = \ \arg\max_{y_1^T} \ P_\theta(x_1^T, y_1^T)$$

HMMs as factor graphs

$$\Psi_t\,(y_{t-1},\ y_t,\ x_t)\ =\ q(y_t|y_{t-1})\ \cdot\ e(x_t|y_t)$$

HMMs can be defined in terms of transitions (factors) $\Psi_t\,(y_{t-1},\ y_t,\ x_t)$, such as:

$$P_\theta(x_1^T, y_1^T)\ \stackrel{\text{def}}{=}\ \prod_{t=1}^{T}\ \Psi_t\,(y_{t-1},\ y_t,\ x_t)$$

➢ Probability of the observation

$$P_\theta(x_1^T)\ =\ \sum_{y_1^T}\ \prod_{t=1}^{T}\ \Psi_t\,(y_{t-1},\ y_t,\ x_t)$$

➢ Probability of best the state sequence

$$\widehat{P}_\theta(x_1^T)\ =\ \max_{y_1^T}\ \prod_{t=1}^{T}\ \Psi_t\,(y_{t-1},\ y_t,\ x_t)$$

➢ Finding the best state sequence

$$\widehat{y}_1^T\ =\ \arg\max_{y_1^T}\ \prod_{t=1}^{T}\ \Psi_t\,(y_{t-1},\ y_t,\ x_t)$$

Viterbi algorithm

➢ **Definition**: The maximum probability that state $y_t = s \in \mathcal{Y}$ is reached at time t by emitting the prefix x_1^t.

$$\gamma_t(s)\ \stackrel{\text{def}}{=}\ \max_{y_1^t;\ y_t=s}\ \prod_{i=1}^{t}\ \Psi_i\,(y_{i-1},\ y_i,\ x_i)\ \equiv\ \max_{y_1^t;\ y_t=s}\ \widehat{P}_\theta(x_1^t,\ y_1^t)$$

➢ **Initialization**: $\forall\ s \in \mathcal{Y}$

$$\gamma_1(s)\ =\ \Psi_1\,(\text{null},\ s,\ x_1)$$

➢ **Recursion**: $\forall\ i = 2 \ldots T;$ and $\forall\ s \in \mathcal{Y}$

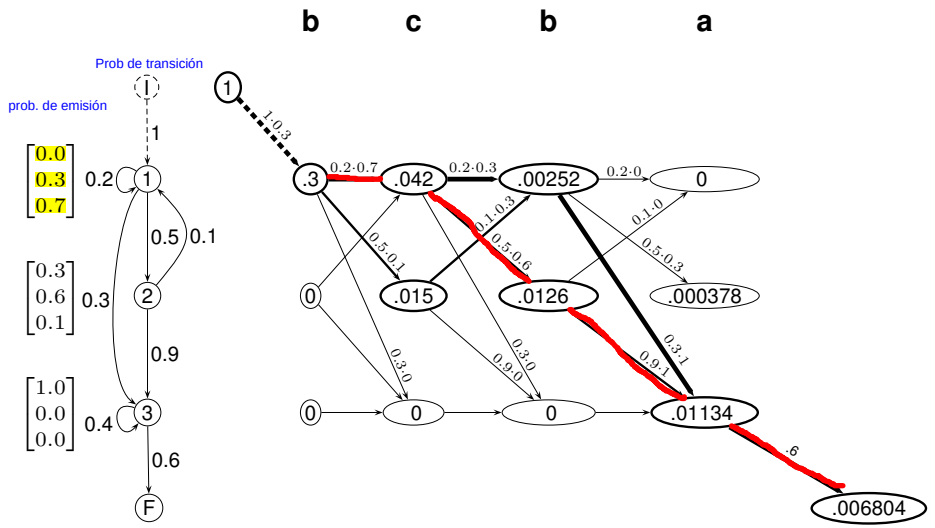
$$\gamma_t(s)\ =\ \max_{s' \in \mathcal{Y}}\ \{\ \gamma_{t-1}(s')\ \cdot\ \Psi_t\,(s',\ s,\ x_t)\ \}$$

➢ **Final result**: The probability of the best state sequence for x_1^T given the model is:

$$\widehat{P}_\theta(x_1^T)\ =\ \max_{s}\ \gamma_T\ (s)$$

Backpointers can be used to recover the optimal sequence $\widehat{y_1^T}$

Algoritmo de Viterbi: ejemplo



Forward algorithm

➢ **Definition**: The probability of generating the prefix x_i^t, reaching the state $y_t = s \in \mathcal{Y}$ at time t.

$$\alpha_t(s)\ \stackrel{\text{def}}{=}\ \sum_{y_1^t;\ y_t=s}\ \prod_{i=1}^{t}\ \Psi_i\,(y_{i-1},\ y_i,\ x_i)\ \equiv\ \sum_{y_1^t;\ y_t=s}\ P_\theta(x_1^t,\ y_1^t)$$

➢ **Initialization**: $\forall\ s \in \mathcal{Y}$

$$\alpha_1(s)\ =\ \Psi_1\,(\text{null},\ s,\ x_1)$$

➢ **Recursion**: $\forall\ i = 2 \ldots T;$ and $\forall\ s \in \mathcal{Y}$

$$\alpha_t(s)\ =\ \sum_{s' \in \mathcal{Y}}\ \alpha_{t-1}(s')\ \cdot\ \Psi_t\,(s',\ s,\ x_t)$$

➢ **Final result**: The probability of the observation x_1^T given the model is:

$$P_\theta(x_1^T)\ =\ \sum_{s}\ \alpha_T\ (s)$$

HMMs as factor graphs

$$\Psi_t\left(y_{t-1},\ y_t,\ x_t\right) \ = \ q(y_t|y_{t-1}) \ \cdot \ e(x_t|y_t)$$

HMMs can be defined in terms of transitions (factors) $\Psi_t\left(y_{t-1},\ y_t,\ x_t\right),$, such as:

$$P_\theta(x_1^T, y_1^T) \ = \ \prod_{t=1}^{T} \ \Psi_t\left(y_{t-1},\ y_t,\ x_t\right)$$

➢ Probability of the observation

$$P_\theta(x_1^T) \ = \ \sum_{y_1^T} \ P_\theta(x_1^T, y_1^T) \ = \ \sum_{y_1^T} \ \prod_{t=1}^{T} \ \Psi_t\left(y_{t-1},\ y_t,\ x_t\right) \ = \ \sum_{s} \ \alpha_T\ (s)$$

➢ Probability of the best state sequence

$$\widehat{P}_\theta(x_1^T) \ = \ \max_{y_1^T} \ P_\theta(x_1^T, y_1^T) \ = \ \max_{y_1^T} \ \prod_{t=1}^{T} \ \Psi_t\left(y_{t-1},\ y_t,\ x_t\right) \ = \ \max_{s} \ \gamma_T\ (s)$$

Probabilistic Context-Free Grammars

Generative models: **Probabilistic Context-Free Grammars**

$$P_\theta(x, t_x) \ = \ \prod_{i=1}^{m} \ p(r_i) \qquad \text{where} \quad t_x : r_1, \ldots, r_m$$

➢ Probability of an observation

$$P_\theta(x) \ = \ \sum_{t_x \in \mathcal{T}_x} \ P_\theta(x, t_x)$$

➢ Probability of the best parse tree

$$\widehat{P}_\theta(x) \ = \ \max_{t_x \in \mathcal{T}_x} \ P_\theta(x, t_x)$$

➢ Finding the best parse

$$\widehat{t}_x \ = \ \arg\max_{t_x \in \mathcal{T}_x} \ P_\theta(x, t_x)$$

Viterbi algorithm

➢ **Definition**: Given $x = x_1 \ldots x_T \in \Sigma^*$ and $A \in N$

$$\widehat{e}(A, i, i+l) \ \stackrel{\text{def}}{=} \ \widehat{P}_\theta(A \stackrel{*}{\Rightarrow} x_{i+1} \ldots x_{i+l})$$

➢ **Initialization**: $\forall\ A \in N; \qquad \forall\ i : 0 \ldots T-1;$

$$\widehat{e}(A, i, i+1) \ = \ p(A \to b) \ \cdot \ \delta(b, x_{i+1})$$

➢ **Recursion**: $\forall\ A \in N; \qquad \forall\ l : 2 \ldots T; \qquad \forall\ i : 0 \ldots T-l;$

$$\widehat{e}(A, i, i+l) \ = \ \max_{B,C \in N} \ \Big\{ \ p(A \to BC) \ \cdot \ \max_{k=1,\ldots,l-1} \ \left\{ \ \widehat{e}(B, i, i+k) \ \cdot \ \widehat{e}(C, i+k, i+l) \ \right\} \Big\}$$

➢ **Final result**: The probability of the best parse tree is:

$$\widehat{P}_\theta(x) \ = \ \widehat{e}(S, 0, T)$$

Backpointers can be used to recover the optimal sequence $\widehat{y}$

Inside algorithm

➢ **Definition**: Given $x = x_1 \ldots x_T \in \Sigma^*$ and $A \in N$

$$e(A, i, i+l) \ \stackrel{\text{def}}{=} \ P_\theta(A \stackrel{*}{\Rightarrow} x_{i+1} \ldots x_{i+l})$$

➢ **Initialization**: $\forall\ A \in N; \qquad \forall\ i : 0 \ldots T-1;$

$$e(A, i, i+1) \ = \ p(A \to b) \ \cdot \ \delta(b, x_{i+1})$$

➢ **Recursion**: $\forall\ A \in N; \qquad \forall\ l : 2 \ldots T; \qquad \forall\ i : 0 \ldots T-l;$

$$e(A, i, i+l) \ = \ \sum_{B,C \in N} \ \Big\{ \ p(A \to BC) \ \cdot \ \sum_{k=1,\ldots,l-1} \ e(B, i, i+k) \ \cdot \ e(C, i+k, i+l) \ \Big\}$$

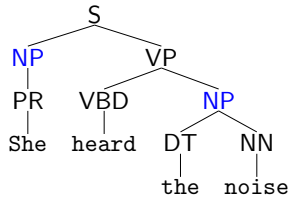
➢ **Final result**: The sentence probability is:

$$P_\theta(x) \ = \ e(S, 0, T)$$

WEAKNESSES OF PCFGS AS PARSING MODELS

- Poor independence assumptions:
  - ⇒ Parsing with PCFG considers conditional independence in the application of the rules

WEAKNESSES OF PCFGS AS PARSING MODELS



- Independence assumptions are often too strong
- Example: the expansion of an NP is highly dependent on the parent (i.e., subjects vs. objects) Statistics for NPs in the Switchboard corpus [Francis et al., 1999]

	Pronoun	**Non-pronoun**
Subject	91 %	9 %
Object	34 %	66 %

Also, the subject and object expansions are correlated !

One way to tackle it: splitting the non-terminals

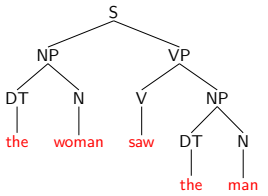
WEAKNESSES OF PCFGS AS PARSING MODELS

- Poor independence assumptions:
  - ⇒ Parsing with PCFG considers conditional independence in the application of the rules
- Lack of lexical conditioning:
  - ⇒ There are lexical dependencies in the syntax analysis (the probability of a syntactic structure can vary depending on the words involved)

WEAKNESSES OF PCFGS AS PARSING MODELS

- Probabilistic CFGs:

$$S \rightarrow NP \;\; VP$$
$$VP \rightarrow V \;\; NP$$
$$NP \rightarrow DT \;\; N$$



The expansion of a verb phrase VP as a verb followed by one or two noun phrases is independent of the choice of verb involved.

- **Example**: [Manning and Schütze, 1999]

	verb			
Local rules	*come*	*take*	*think*	*want*
VP→ V	9.5 %	2.6 %	4.6 %	5.7 %
VP→ V NP	1.1 %	32.1 %	0.2 %	13.9 %
VP→ V PP	34.5 %	3.1 %	7.1 %	0.3 %
VP→ V SBAR	6.6 %	0.3 %	73.0 %	0.2 %
VP→ V S	2.2 %	1.3 %	4.8 %	70.8 %
VP→ V NP S	0.1 %	5.7 %	0.0 %	0.3 %
VP→ V PRT NP	0.3 %	5.8 %	0.0 %	0.0 %
VP→ V PRT PP	6.1 %	1.5 %	0.2 %	0.0 %

One way to tackle it: lexicalized models

WEAKNESSES OF PCFGS AS PARSING MODELS

- Poor independence assumptions:
  - ⇒ Parsing with PCFG considers conditional independence in the application of the rules
- Lack of lexical conditioning:
  - ⇒ There are lexical dependencies in the syntax analysis (the probability of a syntactic structure can vary depending on the words involved)
- It is not considered the context
  - ⇒ The context (eg, slang, knowledge of the subject, etc.) is necessary for proper disambiguation of language